

TASK 7 SUBMISSION

Competitor Demand Prediction Using Job Postings

Demand Forecasting

Forecasting NVIDIA hiring demand

Prepared by	Nayab Khalid
Company specialism	NVIDIA
Task	Task 7 of 10 - Demand Forecasting
Series forecast	Open roles (12 observations) and new postings per day (11)
Horizon	12 weeks, per the shared analysis config
Date	20 September 2026
Repository	github.com/Nayab-khalid/cadetX-Competitor-Demand-Prediction

Series	Model chosen	12 weeks ahead	80% interval
Open roles	Simple exponential smoothing	416	377 to 455
New per day	Mean of last 3	17.7	7.6 to 27.8

The constraint, stated before the results

The shared config sets a 12-week horizon. The NVIDIA series has 12 observations. Forecasting a horizon as long as the entire history is a data problem, not a modelling problem, and no choice of model fixes it. This task therefore fits five simple models and backtests them properly, rather than fitting one complex model that cannot be evaluated. With 12 points, ARIMA order selection or Prophet's seasonality terms would be fitting noise.

1 Method

- **Models:** naive (last value), mean of last 3, drift, simple exponential smoothing, Holt linear trend.
- **Backtest:** expanding origin, one step ahead, over the last 4 observations. Every model is evaluated on data it never saw.
- **Metrics:** MAE and MAPE, both reported, plus bias and residual spread.
- **Intervals:** 80%, built from backtest residual spread widened by the square root of the horizon. They are empirical rather than model-assumed, so they inherit the backtest's small sample.
- **Seasonality:** not modelled. Four months cannot identify an annual cycle, and pretending otherwise would manufacture structure.

The naive baseline is reported next to every model. On a short, flat series it is genuinely hard to beat, and a model that loses to it should be said to have lost.

2 Open roles

Model	MAE	MAPE	Bias	Residual sd
Simple exponential smoothing	5.91	1.38%	-3.14	8.73
Mean of last 3	6.33	1.50%	-0.67	8.52
Holt linear trend	8.45	2.01%	+3.49	8.97
Naive (last value)	9.00	2.14%	+2.00	12.08
Drift	10.96	2.61%	+5.30	12.49

Simple exponential smoothing beats the naive baseline by 34%, from 9.00 to 5.91. That is a real result rather than a formality, and it is what justifies using a model at all here.

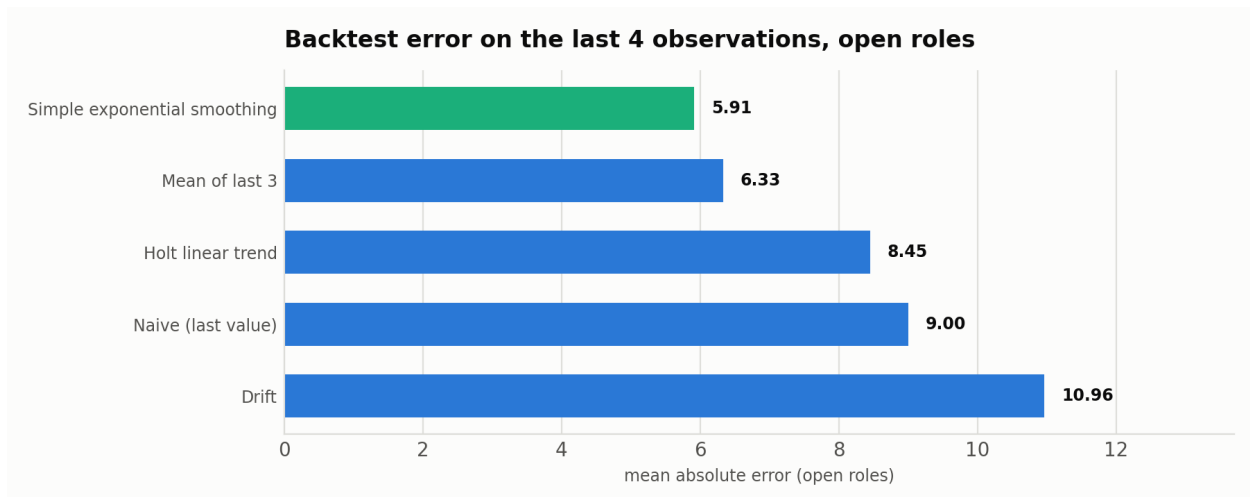


Figure 1. Backtest error on the last four observations.

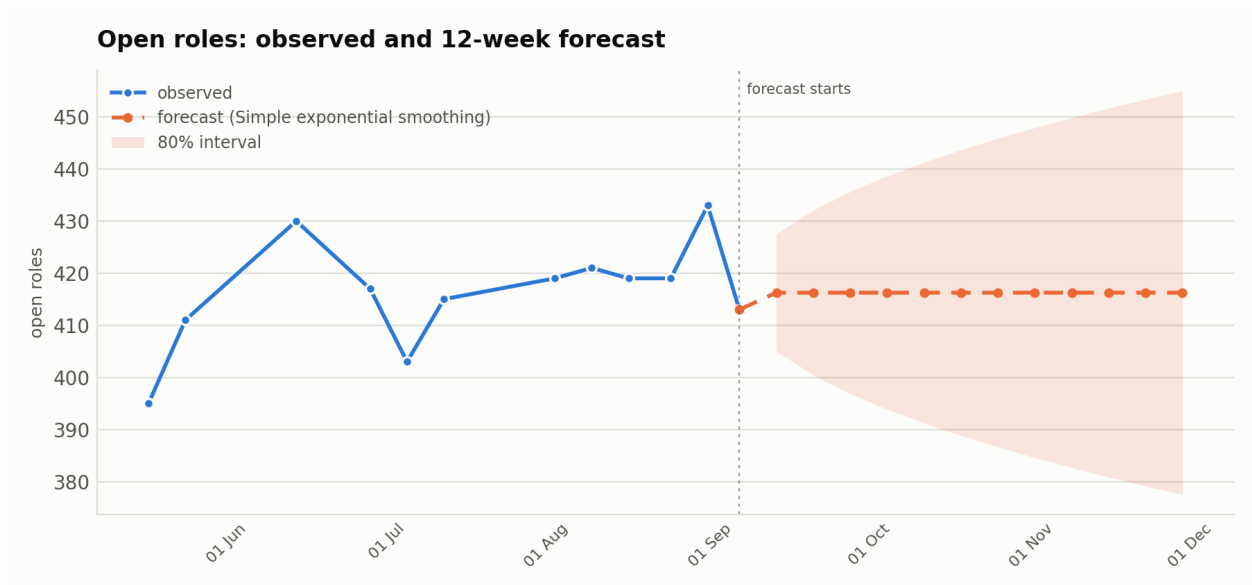


Figure 2. Open roles, observed and forecast, with the 80% interval.

Weeks ahead	Forecast	Lower 80%	Upper 80%	Usable?
1	416	405	427	yes
2	416	400	432	yes
4	416	394	439	yes
12	416	377	455	marginal

At four weeks the interval is 394 to 439, which a planner could use. At twelve it is 377 to 455, wide enough that it only really says "about the same".

3 New postings per day

Model	MAE	MAPE	Bias
Mean of last 3	1.99	10.40%	-1.74
Holt linear trend	2.42	12.97%	-1.17
Simple exponential smoothing	2.78	15.12%	-0.74
Naive (last value)	3.02	16.51%	-0.50
Drift	3.23	17.77%	-0.36

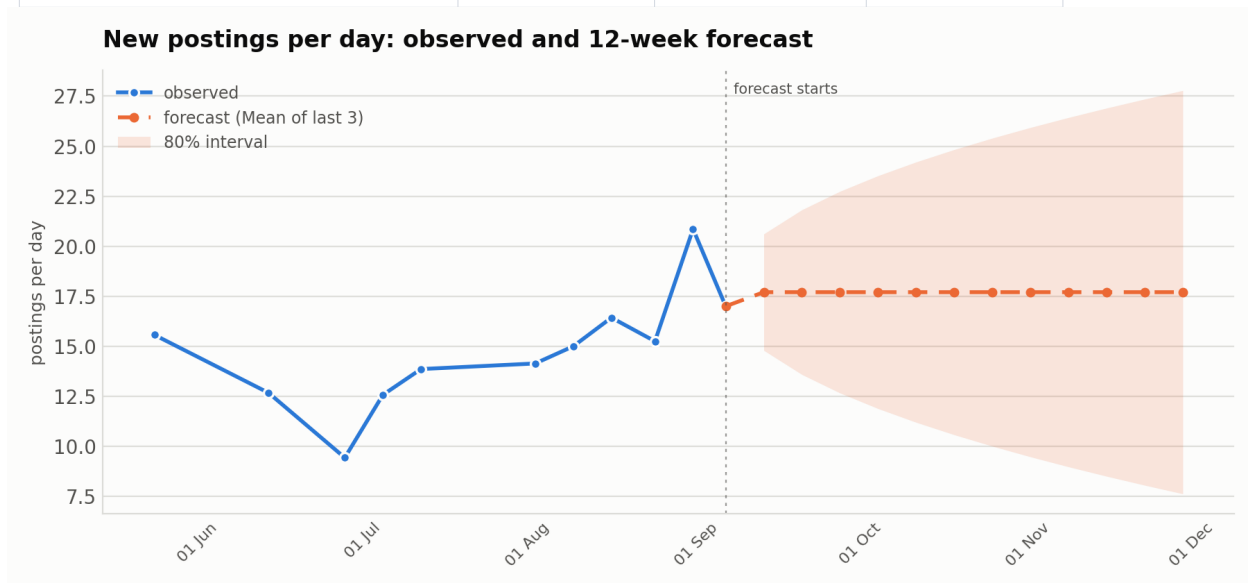


Figure 3. New postings per day, observed and forecast. The 12-week interval spans a factor of 3.6 and carries no information.

The 12-week point forecast is 17.7 per day with an 80% interval of 7.6 to 27.8. That interval is reported because it is what the data supports, not because it is useful. **It is not useful.** Anyone planning against this series should use the 4-week view or wait for more observations.

4 The finding worth reporting

Task 5's upward trend does not forecast

Task 5 measured new-per-day rising at +0.60 per observation, and that is real in the sample. But in the backtest the two models that extrapolate a trend - Drift and Holt - come last and second-last on the flow series, behind a model that assumes no trend at all. The trend is real and it is not projectable. A hiring plan built on extrapolating it would be built on noise. This is exactly what a backtest exists to catch, and it is why the naive baseline is reported beside every model rather than omitted.

Both series forecast essentially flat, which agrees with Task 5: NVIDIA holds roughly 416 open roles and replaces them rather than growing. The forecast is not exciting, and that is the correct answer for this company over this window.

5 Limitations

#	Limitation
1	Twelve observations. Every number here rests on that, and the backtest itself has four points.

#	Limitation
2	The 12-week horizon is not supportable on this data. The 4-week view is the defensible one.
3	Intervals are empirical and inherit the same small sample.
4	Left-censoring in the underlying data means the first observation is not a true arrival count; the flow series starts from the second.
5	No seasonality and no external drivers. Product launches, earnings and hiring freezes are not in the model.
6	This forecasts postings, not headcount. Closed does not mean filled.

6 For the team

1. **Denser collection is the only real fix.** The source has 126 daily snapshots; re-running Task 2 collection at --every 2 would give roughly 60 observations instead of 12, and would improve this task more than any modelling change.
2. **Report the naive baseline.** Any member whose model does not beat last-value-carried-forward should say so plainly.
3. **Three of four companies cannot do this task at all** as their data stands. Google and Microsoft have no usable posting dates and Meta's window is 2024. Task 7 is currently NVIDIA-only, and Task 6 already established that no cross-company time series exists.

Files

File	Contents
data/forecasts/open_roles_backtest.csv	Five models scored on the holdout
data/forecasts/open_roles_forecast.csv	12-week point forecast with 80% interval
data/forecasts/new_per_day_backtest.csv	Same, for the arrival rate
data/forecasts/new_per_day_forecast.csv	Same, for the arrival rate
data/forecasts/forecast_summary.csv	Both series side by side, including the naive baseline
figures/	The three figures above
forecast_demand.py	The analysis, re-runnable from a clean checkout

```
cd work/task-7/nvidia-nayab-khalid
python forecast_demand.py
```

Declaration	No personal or sensitive data is present in any output. No new data was collected; the input is the trend table committed in Task 5.
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Date	20 September 2026